Question:

Tom's age is T years, which is also the sum of the ages of his three children. His age N years ago was twice the sum of their ages then. What is T/N?
$\textbf{(A) } 2 \qquad\textbf{(B) } 3 \qquad\textbf{(C) } 4 \qquad\textbf{(D) } 5 \qquad\textbf{(E) } 6$

Reasoning + Answer:

<think>
1. Let the current ages of the three children be c_1, c_2, and c_3.
2. Given that Tom's age T years ago was twice the sum of the ages of his three children N years ago. Therefore, we can express this relationship mathematically:
\[
T - N = 2(c_1 - N) + 2(c_2 - N) + 2(c_3 - N)
\]

3. Simplify the right-hand side:
\[
T - N = 2(c_1 + c_2 + c_3 - 3N)
\]

4. Rearrange to combine terms:
\[
T - N = 2(c_1 + c_2 + c_3) - 6N
\]

5. Since $T = c_1 + c_2 + c_3$ (the sum of the ages of the three children), we substitute T:
\[
T - N = 2T - 6N
\]

6. Rearrange to isolate terms involving T and N:
\[
T - 2T = -6N + N
\]

7. Simplify further:
\[
-T = -5N
\]

8. Dividing both sides by -1:
\[
T = 5N
\]

9. Therefore, $\frac{T}{N} = 5$.
</think>
<answer>5</answer>